Dhruv Chavan

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EDUCATION

The Ohio State University **Expected Graduation: Spring 2027**
B.S. in Data Analytics - *Honors and Scholars Student* GPA: 3.66/4.0

SKILLS

- **Languages:** C/C++, Python, MATLAB, Linux, Rust
- **Hardware/Platforms:** Arduino, Raspberry Pi, Nvidia Jetson, BNO055 IMU, Marvelmind
- **Tools/Frameworks:** ROS2, OpenCV, GitHub, CMake, Git

EXPERIENCE

Controls and Navigation Research Lead | *Python, MATLAB, C++, ROS2, Arduino* **January 2025 - Present**
Department of Mechanical and Aerospace Engineering at OSU *Columbus, OH*

- Led **5** student team deploying Marvelmind indoor GPS navigation system, achieving **2cm** positioning accuracy
- Implemented Kalman filter from scratch in C++ on BNO055 over I2C, reducing drone orientation error by **85%**
- Designed a 3-DOF IMU test stand with ground truth encoder feedback; accepted at AIAA 2026
- Developing real-time sidestick controller in C++ targeting **≤10ms** latency with **500Hz** force feedback loop
- Producing **5cm** navigation accuracy via robot on Raspberry Pi with automated sensor data pipelines
- Engineering a wildfire payload fusing RGB and infrared sensors with Jetson inference at **30ms** and **92%** accuracy

Data and AI Engineering Intern | *Python, LLMs, Jira, Docker* **September 2025 - December 2025**
BMW Financial Services *Columbus, OH*

- Built MCP server tools connecting internal LLMs to PostgreSQL, improving data retrieval efficiency by **45%**
- Developed **3** Gen AI apps spanning fraud detection, business intelligence, and call center automation
- Launched an AI governance platform enabling **95+** employees to submit model requests for compliance protocols

Undergraduate Student Researcher | *Python, Pandas, Power BI, MySQL, Matplotlib* **July 2024 - Present**
OSU Wexner Medical Center *Columbus, OH*

- Analyzing **4,000+** patient data points; findings presented to clinicians led to a **25%** improvement in outcomes
- Applying hypothesis testing to identify demographic correlations and create visual representations
- Published research as second author at the international RESNA conference and attended the 2025 Rehabweek

Undergraduate Computational Biology Researcher | *Python, R, Pandas* **February 2024 - June 2025**
Nationwide Children's Hospital *Columbus, OH*

- Developed advanced visualizations to analyze brain cancer cell interactions with immune cells
- Analyzed **1.2M+** spatial data points from **30+** tumors, revealing cell clustering linked to ICI response
- Published a paper to *iScience* journal after designing algorithms to analyze tumor structures across cell types

PROJECTS

Autonomous Lego Technic Bugatti | *Python, C++, ROS2, OpenCV, Arduino, Fusion 360* **May 2026**

- Achieved **80%** lane-following accuracy at **30fps** on Raspberry Pi, publishing steering at **20Hz**
- Reduced path deviation by 75% fusing IMU and camera via Extended Kalman Filter on embedded hardware
- Architected **7-node** ROS2 Jazzy stack separating perception, planning, and safety layers mirroring AV stacks
- Engineered safety node stopping the vehicle within **20cm**; achieved **76%** parallel parking success

LEADERSHIP

Mount Scholars Leadership Society | *Member* **August 2023 - Present**

- Led Team Security initiatives and developed leadership through weekly training, mentorship, and retreats
- Completed **75+ hours** mentoring athletes with Down syndrome through weekly tennis sessions and events